

Activity

1. Given

$$P_1 = (5, 4), P_2 = (7, 6)$$

$$C_1 = (2, 4), C_2 = (4, 3)$$

radius Formula $r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

At $C_1 = (2, 4)$

$$r_1 = d(P_1, C_1) = \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9+0} = 3$$

$$r_2 = d(P_2, C_1) = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{25+4} = \sqrt{29} \approx 5.385$$

$\Rightarrow$ Radii : $\{3, 5.385\} \rightarrow 0$ matching values

At $C_2 = (4, 3)$

$$r_1 = d(P_1, C_2) = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{1+1} = \sqrt{2} \approx 1.414$$

$$r_2 = d(P_2, C_2) = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} \approx 4.242$$

$\Rightarrow$ Radii : $\{1.414, 4.242\} \rightarrow 0$ matching values.

Neither
* Therefore * Center gives matching radius values

with just 2 points, with only 2 edge points

both centres gives 2 different radii. In though

voting, neither accumulates votes if forced to

Pick, both have 0 matches.

2. $P(8, 6), C(4, 3)$

$$d = \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{4^2 + 3^2}$$

$$= \sqrt{16+9} = \sqrt{25}$$

$$= 5 //$$

$$\therefore d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{4^2 + 3^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2} = \sqrt{16+9} = \sqrt{25} = 5 //$$

$$(4, 3) = (x_1, y_1)$$

3. $P(-8, 4), C(5, 8)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(8-5)^2 + (4-(-8))^2}$$

$$= \sqrt{3^2 + 4^2} = \sqrt{9+16}$$

$$= \sqrt{25}$$

$$= 5 //$$

$$= \sqrt{(8-5)^2 + (4-(-8))^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16}$$

$$= \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5 //$$

$$= \sqrt{(8-5)^2 + (4-(-8))^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16}$$

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$$= \sqrt{(8-5)^2 + (4-(-8))^2} = \sqrt{3^2 + 4^2} = \sqrt{9+16}$$